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Social implications of human food subsidies on wildlife populations

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Human activities generate a variety of novel food sources that wild animals exploit. On land and in water, these human-derived foods can profoundly alter intraspecific interactions in wildlife with cascading effects on population dynamics and ecosystem functioning. Yet, despite their growing ecological relevance, the role of human food subsidies in shaping intraspecific interactions remains underexplored. We propose a novel framework that captures how key characteristics of human food subsidies—such as high abundance, predictability, increased proximity to humans and dietary composition—modify social interactions. Specifically, we discuss how individual-level changes in fitness, time allocation, movement and social choices can shape interactions and the resulting social network properties, both directly and indirectly through alterations in population size, density and composition. Finally, we outline the ecological and evolutionary consequences of these social changes, including impacts on the transmission of diseases, stress and information, as well as on selection and development. Collectively, these alterations highlight the broad social implications that intentional and unintentional human food subsidies can have for ecological and evolutionary processes in wildlife populations.

1. Introduction

The Anthropocene is marked by rapid human-driven changes that are reshaping terrestrial and aquatic ecosystems [1–4] with profound implications for wildlife behaviour, including intraspecific interactions [5–8]. Social interactions, in particular, are fundamental to animal life, and their frequency and organization give rise to a population's social structure [9,10]. The positioning of individuals within the social structure can influence their survival [11,12] and reproductive success [13,14], both of which are critical determinants of fitness. At the group and population level, social structure governs key processes such as the transmission of diseases [15,16] and information [15], dispersal [17,18], mating systems [19,20] and modulates how selection acts on phenotypic traits [21,22]. Hence, understanding how human-induced environmental changes modify social structure is crucial to anticipate the consequences for future population functioning and viability.

Animal social structure emerges from multiple interacting factors. These include population-level characteristics, such as population size and density [23,24], phenotypic composition [25,26], as well as individual-level decisions with whom and how frequently to interact [9,27]. Increasing evidence shows that human activities—including pollution, direct human presence and climate change—can alter these factors, with cascading effects on

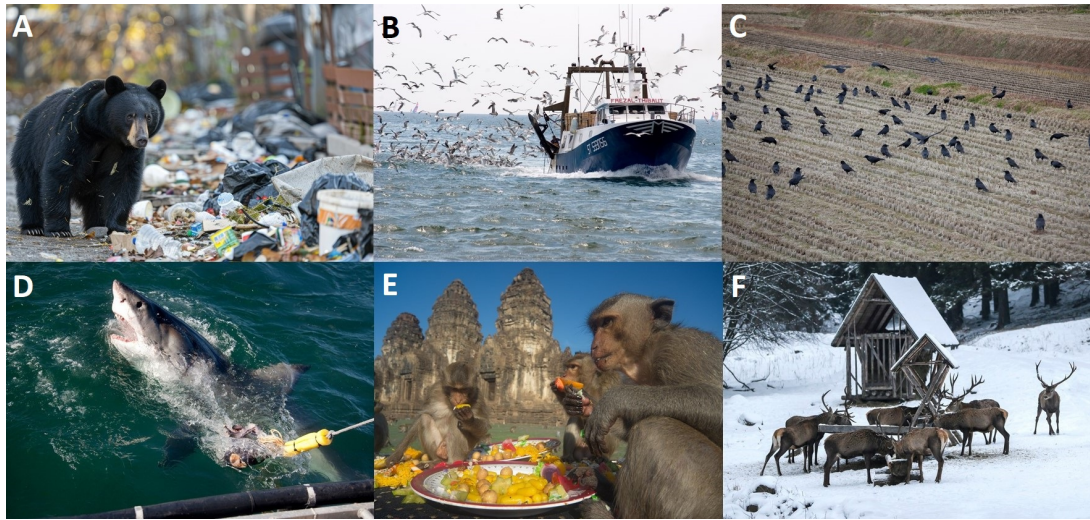


Figure 1. Examples of intentional and unintentional human food subsidies. (A) Bear scavenging from waste. (B) Seabirds following fishing trawlers to feed on discarded fish. (C) Crows foraging in a crop field. (D) Shark attracted with meat during cage-diving. (E) Macaques consuming food offerings at a temple. (F) Supplemental feeding of deer during winter. Photo credits: (A) Serhii/Adobe Stock, (B) Simon Ebel/Adobe Stock, (C) Paylessimages/Adobe Stock, (F) Conchi Martinez/Adobe Stock, (E) topten22photo/Adobe Stock, (D) Michal/Adobe Stock.

population-level social structure [7,28–30]. However, a so far often-overlooked human activity with considerable consequences for intraspecific interactions is human food subsidies.

Human modifications to food availability date back to the late Palaeolithic, when nomadic hunter–gatherer societies unintentionally provided scavenging species with food remains such as carcasses [31–33]. For example, synanthropic behaviour in small carnivores such as red and Arctic foxes (*Vulpes vulpes* and *V. lagopus*) emerged as early as 42 000 years ago [33]. As such, food subsidies may be one of the earliest forms of human-induced environmental change, with lasting effects on animal behaviour and ecosystems that persist until the modern day. In the present, human-generated food sources are ubiquitous and exploited by many animal species (box 1; figure 1). These subsidies include all food made available through human activities (see [34]), whether intentional—such as feeding stations, religious offerings, or wildlife feeding by tourists—or unintentional, arising from agriculture (e.g. crops, livestock), food waste (e.g. landfills, rubbish bins, restaurants) and hunting or fishery discards (box 1).

Box 1. Intentional and unintentional human food subsidies.

Humans provide a vast range of food sources to wildlife (figure 1). For example, it is estimated that 30–40% of all food is wasted globally [34], with landfills accumulating waste at rates outpacing urbanization [35]. Landfills are often highly abundant and predictable in both space and time, thus constituting a highly profitable food source that is exploited by various species, including mammals, birds, amphibians and reptiles [36]. Similarly, waste bins and food leftovers at restaurants are an attractive food source to many terrestrial animals [37]. Fishery and hunting discards, as well as middens, are another globally abundant human-derived food source. Approximately 8% of caught fish is discarded [34,38] and up to 10^5 tonnes of carcasses are left in the field by game hunters in the United States each year [34]. Additionally, agricultural land covers about 36% of the world's land area (<http://data.worldbank.org/>) and provides crop and residues that can be used as a food source by wildlife [34,39], while roads can generate carrion that benefits scavenging species [40]. Humans also intentionally provide food to animals for various reasons. Feeding stations for birds and game, for example, are common practices. In the United Kingdom, 50% of households with garden access provide supplementary food for birds [41], and supplemental feeding of game during winter is often used for wildlife management and hunting purposes [42]. Feeding of wild animals has also become a popular tourist activity. For instance, managed tourism programmes frequently provide food to dolphins and sharks to facilitate close encounters [43,44]. Finally, feeding animals plays a significant role in some religious practices. In Indian culture, people commonly feed animals across all taxa, either routinely or during festive occasions, often at sites of religious significance such as temples [45].

Importantly, human food sources differ markedly from natural food sources in various ways. They are often highly abundant, predictable in space and time, distinct in nutritional composition and expose animals to novel stimuli such as traffic, noise and direct human presence [34]. These features can influence individuals by altering fitness, movement decisions, time allocation across behaviours and social choices (i.e. with whom and how to interact). Such individual-level changes can shape social interactions directly, through altered social decision-making, and indirectly, through their effects on population size, density and composition. These pathways can, in turn, drive shifts in social network properties, including network density, fragmentation and connection strength. Yet, despite recognition of the widespread ecological impacts of human food subsidies [34], the social implications of food subsidies on wildlife populations remain poorly known.

Here, we outline the pathways through which human food subsidies can alter intraspecific interactions and, consequently, wildlife population dynamics. We begin by examining the social implications of human food subsidies, addressing both their direct and indirect effects on social interactions. Specifically, we discuss how key features of human-generated foods—such as their abundance, predictability, proximity to humans and dietary composition—shape social interactions and population social structure through their effects on individual-level traits (e.g. survival, movement). We then discuss how these changes in social structure can impact population-level processes, including the transmission of diseases and the direction and intensity of selection. While human food sources also impact intraspecific interactions in invertebrates, we here focus on vertebrates for which more research is available from wild populations rather than captive systems. By providing a conceptual framework (figure 2), we aim to stimulate future research and deepen our understanding of the diverse impacts of human activities on wildlife populations.

2. Social implications of human food subsidies

(a) Direct effects

Human food sources can directly impact the quantity and quality of social interactions through changes in individual social decision-making processes (i.e. with whom and how to interact, figure 2(i)). For example, many human subsidies differ markedly from natural diets in nutrient composition, with implications for social behaviour. Because foods high in sugars and fats are perceived as high-value resources, they can directly intensify competition and aggression by elevating resource value [46] and through physiological mechanisms [47]. Although links between diet composition and aggression are well established in humans [48], corresponding research in non-human animals remains limited. For example, dogs on high-protein diets were significantly more likely to show aggressive behaviours towards both their owners and other dogs [49], and high-fat diets increased aggression levels in captive male rats and mice [47]. Finally, human food can be heavily contaminated with pollutants such as heavy metals [50], which may impair the neurological and physiological functions that underlie communication, coordination and other social behaviours [30]. However, studies examining these patterns in wild animals—and the underlying physiological mechanisms—are notably scarce.

In addition, social behaviour is often highly plastic, enabling individuals to adjust their associations and flexibly shift between solitary and group living in response to ecological conditions. In many species, for example, social foraging emerges as an adaptive strategy that enhances foraging efficiency in unpredictable environments [51,52] and offers protection from predators [51,53]. Human food sources can fundamentally reshape ecological conditions—specifically, the spatial and temporal distribution of food and risk—thereby influencing social decision-making and time allocation, with effects that can scale up to alter animal social structure.

Human food sources can diminish social interactions when their high abundance and predictability reduce both the benefits and necessity of social foraging. Under such conditions, individuals may choose to forage alone or within smaller, cohesive subgroups of preferred associates. These shifts can lead to more fragmented social structures, characterized by smaller and more modular groups at the population level. For example, chacma baboons (*Papio ursinus*) were more dispersed and less cohesive when foraging in urban environments, likely reflecting the widespread availability of food resources [54]. Similarly, bottlenose dolphins that frequently begged for food from humans were less socially connected, probably because human-provided prey can be obtained individually rather than cooperatively, isolating these individuals from conspecifics that forage on natural prey [55].

Beyond changes in food availability, human-induced disturbances—such as noise, pollution and the presence of humans [56]—along with the challenges of navigating altered habitats, can further constrain social behaviour by imposing time limitations. Animals may need to travel longer distances to reach provisioned sites or spend time monitoring human activity before approaching [54,57]. In two macaque species (*Macaca mulatta*; *M. radiata*), individuals that frequently interacted with humans spent significantly less time resting and groomed fewer group members [58,59]; and food-provisioned female bottlenose dolphins spent less time socializing [60]. Such patterns may arise because provisioned individuals must invest time in atypical foraging strategies—waiting for safe opportunities to approach humans or handling difficult-to-access food items, such as opening rubbish bins.

Human food subsidies can also foster social interactions if they reduce competition and conflict among group members [61], free up time that would otherwise be spent foraging [62] or increase perceived predation risk [56]. For instance, food-supplemented vervet monkeys (*Chlorocebus pygerythrus*) spent more time resting and socializing, and groomed conspecifics more frequently, compared to periods before or after supplementation [62]. In addition, the presence of humans near food sources can act as a ‘shield’ against predators or competitors (the *human shield effect* [63]), allowing animals to reallocate time towards other behaviours such as social interactions [64]. Highly urbanized fox squirrels (*Sciurus niger*), for example, exhibit reduced vigilance and weaker responses to predator vocalizations than less urbanized individuals [65], potentially freeing time for other activities. Finally, human subsidies can also encourage grouping and social contact as a coping strategy to mitigate stress or enhance safety. In long-tailed macaques (*Macaca fascicularis*), individuals at sites with high human activity engaged in more frequent grooming, suggesting that grooming served as a social mechanism to reduce experienced stress caused by human presence [66]. While these examples are not directly caused by human food provisioning, similar effects on vigilance and social behaviour could plausibly occur at sites where human food subsidies are present. Determining whether human food subsidies ultimately free up time or create time constraints—and how these outcomes influence social interactions across various species—remains an important avenue for future research.

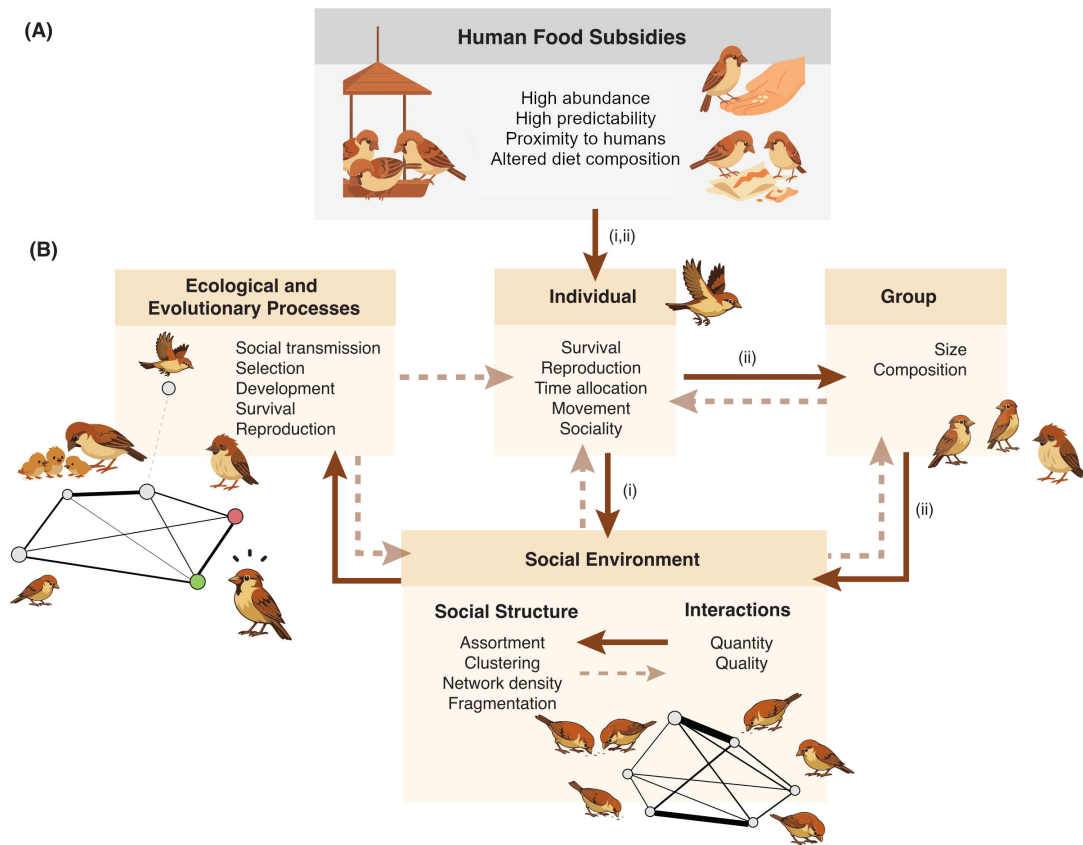


Figure 2. Direct and indirect effects of human food subsidies on animal social structure and their implications for ecological and evolutionary processes. Human food subsidies are characterized by increased abundance, predictability, proximity to humans and altered diet composition (A), which generate cascading effects on individual behaviours and social dynamics (B). By influencing individual social decisions (i.e. sociality), human subsidies can directly (i) modify social interactions and the resulting properties of social structure (e.g. assortment, fragmentation). In addition, individual-level changes—such as altered movement decisions and survival—can indirectly (ii) shape social interactions by affecting the size and composition of animal groups and populations. These shifts in the social environment can, in turn, influence ecological and evolutionary processes, including the social transmission of pathogens, stress and information. Feedback loops (dashed arrows) from these processes to social structure, and to group- and individual-level dynamics [10], exist but are not elaborated on in this review.

(b) Indirect effects

Human food sources can indirectly shape social interactions and the emergent properties of animal social structure by altering the size, density and composition of groups and populations (figure 2(ii)). These effects arise from changes in individual movement decisions (i.e. when and where to move), which determine opportunities for social encounters [67], as well as from shifts in survival and reproduction [34] that influence the future number and phenotypic composition of social actors.

Increasing evidence shows that individuals across various taxa reduce their movements and space use in response to human food subsidies [68–71]. This may be due to the predictable, abundant and highly profitable nature of human food, reducing the need to range widely in search of resources. In Yellowstone National Park, United States, grizzly bears (*Ursus arctos horribilis*) fed almost exclusively from waste disposal sites; once these were no longer available, bear densities declined rapidly and individuals massively enlarged their spatial range [72]. Similarly, four-striped grass mouse (*Rhabdomys pumilio*) females reduced their home ranges by 43% when being supplemented with food [73]. At local scales, such reductions in movement and space use can increase population density by concentrating individuals in limited areas (e.g. landfills), thereby heightening the frequency of social encounters [74,75]. These effects may be further amplified when human subsidies occur as short-lived, predictable resource pulses, which temporally concentrate individuals, intensify social interactions and potentially increase competition or aggression. For example, food provisioning associated with tourism is often limited to certain, recurrent times of day (e.g. lunchtime) or specific seasons (e.g. holidays). In Shark Bay, Australia, bottlenose dolphins are fed with fish three times daily between 7.30 and 13.00 [44]. Similarly, several bird species, including pigeons and gulls, have been observed gathering around noon at sites where people regularly eat lunch [76,77]. Sulfur-crested cockatoos (*Cacatua galerita*) visited balconies (to receive food) before and after working hours when residents were most likely to be home [78], and feeding of chimpanzees (*Pan troglodytes*) in Gombe National Park, Africa, led to heightened aggression, as profitable food was available only briefly during a short time [79].

Although human food can increase social encounter rates at local scales, it may decrease them at broader spatial scales. Reduced movement and space use can limit encounters among individuals and thereby diminish connectivity within and between populations. For example, some migrating species that utilize human food are increasingly reported to cease migration altogether, forming stationary populations [80,81] or shorten their migration routes [82], fundamentally reducing long-distance

movements and population connectivity. Thus, the consequences of altered animal movement for intraspecific interactions may depend on the spatial scale of movement, highlighting the need for further research.

In addition, differences in individual phenotypes mediate the use of human food subsidies, thereby shaping movement patterns and, consequently, the local composition of populations. Sex- and age-biased foraging is increasingly reported across taxa. In primates, for instance, males more frequently associate with humans and utilize human-provided food (e.g. chimpanzees [83], moor macaques (*Macaca maura*) [84], chacma baboons [57]). In contrast, younger individuals more often exploit human food in bearded vultures (*Gypaetus barbatus*) [85], while older individuals do so in white storks (*Ciconia ciconia*) [86]. Such non-random use of human food by particular phenotypes can shape patterns of association: individuals with similar responses to human subsidies are more likely to encounter and interact with one another, leading to phenotypically structured groups (i.e. network assortment). In toothed whales (e.g. orcas (*Orcinus orca*), sperm whales (*Physeter macrocephalus*) and bottlenose dolphins (*Tursiops* spp.)), social community membership is often linked to foraging responses such as depredation, scavenging, cooperation or begging in association with various fishing activities (e.g. long-lining, trawling, net-casting and recreational fishing) [87]. Individuals that forage in association with humans are then more likely to associate with one another [88] and are part of the same social community [89,90], whereas the removal of the human food source (e.g. prawn trawler discards) reverses changes in (dolphin) community membership [91].

Human food subsidies can also influence social interactions through their effects on individual survival and reproduction [34]. The high abundance and predictability of supplemental food often enhance survival [92,93] and reproductive success [94–96], thereby increasing population size—a key determinant of social dynamics. Larger populations can elevate density, group size [97–99] and social encounter rates [24] while also influencing various measures of social structure such as network centrality, fragmentation, the density of social connections and the strength of social relationships [23,100,101]. However, supplemental feeding can also have no or negative effects on survival and reproduction if the supplemental food is of poorer quality than natural food [102], if immunity is compromised [103], or if disease transmission is facilitated [104].

By altering individual fitness, human food subsidies can also modify the phenotypic and demographic composition of populations, thereby reshaping social association patterns. For example, in winter, adult tit species (*Paridae*) generally have higher survival than yearlings, likely due to greater foraging experience and dominance [105]. Increased adult survival through human-provided food would result in an older age structure, maintaining more experienced individuals within the population. In contrast, human food subsidies can also enhance reproductive success and juvenile survival, thereby offsetting natural mortality biases and shifting population age structure towards younger individuals [105]. Younger animals often maintain more and broader social connections compared to older, more selective conspecifics [106]. Therefore, the demographic consequences of human food subsidies—whether promoting adult survival or juvenile recruitment—can differentially shape social networks, potentially leading to either more stable, age-structured systems or denser, highly interconnected social structures.

3. Ecological and evolutionary consequences

Numerous studies have documented the effects of human food subsidies on population processes such as disease prevalence, animal migration and natural selection [34,81,107,108]. However, the mechanisms linking human food subsidies to these ecological and evolutionary outcomes often remain unclear. In this section, we explore how such effects may arise through changes in intraspecific interactions (figure 2). We synthesize empirical evidence to argue that examining the link between human food subsidies and social behaviour can provide new insights into social transmission, development, as well as selection and the evolution of cognitive traits. Developing a deeper understanding of the social implications of human food subsidies, along with their ecological and evolutionary consequences, will offer valuable information to predict the consequences of future environmental changes, such as the removal of subsidies (e.g. landfill closures) or increased food availability due to tourism.

(a) Social transmission

(i) Disease transmission

Human food subsidies are frequently linked to higher disease transmission. This is largely because human subsidies amplify exposure to pathogens and facilitate spread through increased social encounters [109]. For instance, wild birds show a higher prevalence of infectious diseases among individuals foraging at supplemented forest sites compared to those at unsupplemented sites [110], and experimentally increased feeder density led to a higher pathogen transmission in captive house finches (*Haemorrhous mexicanus*) [111]. Human food subsidies can also heighten disease transmission by directly altering the type of social interactions. For example, banded mongooses (*Mungos mungo*) were more aggressive when foraging from garbage compared to natural habitats, leading to more injuries and greater tuberculosis prevalence [112]. Although many studies point to changes in intraspecific interactions as the mechanism driving increased disease prevalence at supplemented sites, few explore the number and strength of social connections between individuals.

While pathogen spread is highly context-dependent and can vary across species, pathogens and ecological conditions, mapping animal social networks often suggests that greater clustering (well-connected subgroups) and stronger connections can facilitate disease transmission, whereas high modularity (i.e. social fragmentation) may reduce it [113,114]. Studies examining disease transmission in the context of human subsidies typically focus on small spatial scales (e.g. one local population), but it would be valuable to gather detailed information on the use of human food sources, intraspecific interactions and disease transmission at larger spatial scales (i.e. across local populations). This broader perspective is important because

human food subsidies may increase social encounter rates locally while simultaneously reducing encounters at larger spatial scales (see §2b). Such shifts in movement and contact patterns can fragment social communities, reduce interactions between populations and potentially decrease the wider spread of pathogens. For instance, migration is often assumed to facilitate the geographical spread of pathogens, as animals are exposed to diverse parasites while moving annually between breeding and wintering grounds [115,116]. However, species that utilize human food sources are increasingly reported to cease migration altogether, forming stationary populations [80,81], while others shorten their migration routes [82]. By reducing long-distance movements—coupled with findings of decreased movement in human-modified environments [117]—human food provisioning may thus limit disease spread on a broader scale.

Consequently, the influence of food subsidies on disease outcomes may be a question of scale, with patterns observed locally not necessarily aligning with those at broader scales, warranting future research to determine whether these relationships hold. To better understand these dynamics, studies conducted at larger spatial scales, along with experimental manipulations of access to human food subsidies, can provide critical insight into how resource availability shapes social interactions, pathogen transmission and opportunities for disease mitigation. During outbreaks, wildlife authorities often recommend halting supplemental feeding. Yet animals commonly respond to sudden resource loss by expanding their range [72,118]. The removal of human food sources may therefore reduce local social contact but increase wider-ranging movements that bring previously isolated groups into contact, potentially facilitating disease spread at larger spatial scales.

(ii) Stress transmission

While direct effects of food abundance, predictability and altered predation risk can influence physiological stress, human food-associated changes in social interactions can further shape how stress emerges and spreads within populations. Individuals that interact with stressed conspecifics—whether the stress is acute or chronic—often exhibit similar physiological responses (e.g. elevated circulating stress hormones) and behavioural shifts (e.g. altered movement patterns), consistent with the transmission of stress [119,120]. At human food subsidies, intensified agonistic interactions such as competition and aggression [47,79,112] may promote stress, while increased social connectivity can likewise facilitate the spread of physiological states across individuals, analogous to pathogen transmission. Thus, for species that access human food subsidies, stress transmission likely plays an important ecological role, emerging from both the direct physiological effects of supplemental feeding and its impacts on social connections.

(iii) Information transmission

The structure of social networks also influences the spread of information and novel behaviours across populations [10,15]. In contrast to disease transmission, individuals can actively decide how to respond to newly acquired information. For instance, an individual may only adopt a novel behaviour if it is demonstrated by the majority of the group or by specific, influential members [121]. Consequently, while high network density may facilitate pathogen transmission, it does not necessarily increase the spread of information. Moreover, abundant and predictable human food subsidies can directly reduce the value of social information [52,122], thereby influencing both the production of and responses to new information. For instance, bat species that feed on unpredictable prey are more likely to forage near conspecifics and emit echolocation calls indicating prey capture, whereas this behaviour is less common among species that target predictable food sources [123]. Human subsidies may therefore shape the diffusion of behaviours in two ways: by altering social network connectivity and by modifying individual decision-making processes that govern how information is produced, shared and acted upon.

(b) Socio-cognitive traits

The evolution of cognition is hypothesized to be partly driven by the demands of group living (the social intelligence hypothesis) and ecological challenges such as finding and accessing food [124]. Human food subsidies can alter these ecological challenges by easing food acquisition while introducing new ones that require advanced cognitive skills—such as interpreting human behavioural cues to locate food [125,126] or remembering the timing and location of food [76–78]. At the same time, because subsidies modify the social environment, they can also influence the cognitive demands of group living. For example, subsidies can increase group sizes by locally concentrating individuals. Group size has been extensively examined in relation to cognitive performance, with larger groups often associated with enhanced problem-solving abilities and increased behavioural innovation [127]. One explanation for this phenomenon is the ‘pool of competence’ effect, which suggests that larger groups contain individuals with diverse skills, increasing the likelihood of problem-solving [128]. Additionally, increased competition in larger groups may drive improved problem-solving and behavioural innovation, a concept known as the ‘skill pool effect’ [129]. For example, in vervet monkeys, individuals facing higher feeding competition learnt a new and more efficient foraging skill faster [130], and in three-spine sticklebacks (*Gasterosteus aculeatus*), high intraspecific competition led to greater diversity in food resources used [131]. Therefore, human food subsidies may enhance problem-solving performance and behavioural innovation through increases in group size.

In addition, changes in group size and social structure can increase the cognitive demands of navigating social relationships. Living in larger or more dynamic groups requires individuals to remember more social partners, track dominance hierarchies and manage both affiliative and competitive interactions. Such social complexity is thought to select for advanced cognitive abilities, including individual recognition, memory and third-party relationship understanding [132–134]. Together, these

dynamics suggest that human food subsidies not only reshape the ecological challenges animals face but may also alter the cognitive landscape of their social lives, with potential consequences for cognitive development and evolution.

(c) Development

Human food can affect not only the individuals directly receiving the subsidy but also their offspring, with downstream developmental effects on sociality. Having more food is then likely to have impacts on the growth and traits of the offspring. For example, red kite (*Milvus milvus*) chicks that received food supplementation (via their parents) developed distinctly different social and spatial preferences from those that did not [135]. The youngest (and least competitive) chick in the nest typically avoids conspecifics after hatching and during dispersal. However, in nests where parents received food supplements, the youngest chick instead became the most social after fledging [135]. These effects are likely tied to physiological conditions experienced during development [136], which are very likely to be altered by food supplementation. Such effects could be direct (e.g. via the amount of food that young animals receive during development [137]) or indirect (e.g. via stress experienced by their parents; see also §3a(ii)). The links between food supplementation, development and future social behaviour are poorly understood and require urgent study.

(d) Selection

While it is recognized that human food subsidies influence selection pressures and drive evolutionary changes in natural populations [108,138], the impacts of subsidy-driven shifts in intraspecific interactions on population and evolutionary dynamics remain largely overlooked. Simply being part of a group has fitness consequences, such as the benefits of social foraging or the costs of increased competition [139]. Human-provided food can modify these effects by influencing group size, elevating competition and disease prevalence, enabling rapid transmission of stress and potentially reducing the fitness benefits of social foraging when food is abundant.

Fitness and selection consequences also depend on the phenotypic traits of social associates. In animals that exhibit social behaviour, selection can be mediated by the interaction between an individual's phenotype and those of its associates—a concept known as social selection [140]. For instance, if a certain trait, such as body size, increases the chances for copulation, the individual's reproductive success may be influenced by the average expression of this trait among its associates. The resultant effects on phenotypic selection emerge at the population level when social selection gradients are experienced nonrandomly among individuals (i.e. positive or negative phenotypic assortment [141,142]). Thus, when populations become phenotypically structured, local differences in the social environment that individuals experience can influence the strength and direction of selection [22]. Because human food sources can cause positive phenotypic assortment, such as if individuals of similar phenotypes preferably associate, it is expected that food supplementation impacts selection.

Individuals can also adjust their phenotype in response to the conspecifics they associate with (i.e. social plasticity [143,144]). For example, individuals might exhibit higher levels of aggression in the presence of more aggressive conspecifics compared to the presence of more passive ones [145]. Consequently, indirect genetic effects (i.e. when one individual's genes affect another individual's phenotype [146]) can alter selection by either decreasing or increasing genetic variance, affecting evolutionary trajectories. Human food subsidies may play a role here if they attract individuals with particular traits—for example, more aggressive individuals—leading to changes in phenotypic variation in the population and selection strength. The effect of social plasticity on evolutionary outcomes further depends on the pattern of social interactions within a population. For example, the contribution of indirect genetic effects to phenotypic variance can be highest at intermediate social network densities and decreases at higher densities [22]. This is because, in highly connected social networks, the social environment becomes homogenized across individuals [22]. By locally concentrating individuals and thus increasing network densities, human subsidies may reduce the impact of indirect genetic effects.

In addition, human food subsidies may influence frequency- and density-dependent selection. When food is scarce, alternative behavioural phenotypes may each gain a fitness advantage when rare, a phenomenon known as negative frequency-dependent selection [147,148], maintaining behavioural diversity. By increasing food availability, human subsidies can reduce the advantages of rare behavioural types, altering social interactions, competitive outcomes and decreasing behavioural diversity within populations. Similarly, by locally concentrating individuals, human subsidies may intensify competition and modify the strength and direction of density-dependent selection, further shaping evolutionary trajectories.

(e) Interspecific interactions

Although we primarily focus on intraspecific interactions, many of the social implications of human food subsidies extend to interspecific interactions, with potential consequences for community structure and ecosystem processes [34,149]. Changes in population size and density resulting from food supplementation can alter interactions not only within, but also across, trophic levels among supplemented and unsupplemented species [150,151]. For example, human food subsidies attract pigeons into urban areas, where they constitute one of the primary food sources for raptors such as peregrine falcons (*Falco peregrinus*) [152,153]. During the COVID-19 pandemic and resulting lockdowns, however, falcons substantially shifted their diets, consuming far fewer pigeons—likely because pigeons abandoned city centres and foraged in the countryside in response to the absence of human-provided food [153].

Human food subsidies also strongly shape interactions between humans and wildlife. While feeding wildlife can provide psychological and cultural benefits, contributing to human wellbeing [154,155], it is also a major driver of human–wildlife conflict. Food subsidies can lead to habituation and reduced wariness in wildlife, while crop and property damage often heighten aggression on both sides [156]. Moreover, food-induced increases in wildlife population size and density can intensify human–animal encounters, and associated shifts in phenotypic composition may further accelerate conflict. For instance, in brown bears (*Ursus arctos*), females and young individuals are more frequently observed near human settlements [157]. If available food subsidies also enhance the survival of these individuals, such patterns may become further amplified, leading to a growing number of bears regularly inhabiting areas close to humans. Such patterns could then be further reinforced through cultural transmission if young bears learn from their mothers and other conspecifics to exploit human food subsidies. The complex interplay between food subsidies and wildlife social dynamics underscores the importance of future research to better understand these interactions and develop strategies that mitigate conflict while promoting coexistence.

4. Conclusion

Human food subsidies are more than ecological inputs; they are transformative forces reshaping the social dynamics of wildlife populations. By altering the abundance, predictability, proximity to humans and nutritional composition of resources, subsidies influence fitness, how animals allocate time, where they move and with whom they interact. These direct and indirect effects on intraspecific interactions (figure 2(i),(ii)) can cascade from individuals to groups, restructuring social systems with profound ecological and evolutionary consequences including social transmission, development, selection and even broader community dynamics. Crucially, the outcomes are not uniform. They vary with the ecology of the species, the type of food source and animals' perceptions of humans. The same subsidy may suppress social opportunities in one context while amplifying them in another, with impacts ranging from heightened local disease risks to reduced connectivity across populations. Recognizing food subsidies as both ecological and social drivers offers a new perspective on anthropogenic change. The central challenge now is to move beyond documenting effects and towards developing predictive principles: under what conditions do human food subsidies destabilize animal societies, and when might they instead foster novel cooperation, competition or transmission pathways? Addressing these questions will be essential for understanding how human-provided subsidies are shaping the future of animal societies in the Anthropocene.

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Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. K.B.B.: conceptualization, visualization, writing—original draft, writing—review and editing; M.C.: conceptualization, writing—review and editing; D.R.F.: conceptualization, writing—review and editing; T.M.: conceptualization, writing—review and editing. All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

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